

Hearth OS

Progress report · version 0.23.0 · 30 September 2026

Hearth OS turns a living-room PC into a TV-style home screen for Steam, emulation, streaming apps and Quick Resume, driven by a controller, a TV remote or a Wii Remote. It's built on Bazzite and installs as a system image that updates itself and can always be rolled back. It runs on AMD and Intel graphics; NVIDIA is left for later.

17,261

lines of code

458

automated tests

26

versions released

27

pull requests merged

15

guides in docs/

Since 0.22.4: code +1,776, tests +77. Tested on: Ryzen 7 5800X3D, Radeon RX 6750 XT, 24 GB, 256 GB SATA SSD, Xbox controller, Wii Remotes on a DolphinBar.

What's new in 0.23.0

- **Pictures on the tiles:** Settings is a gearbox, Sleep a moon, Restart a lap of a circuit, Power Off an engine start button, and so on. Apps from Flathub show their own icon.
- **Fixes from the first field tests on the PC:**
 - Steam: when a game started from its tile ends, Steam closes and you're back home, instead of stuck in Steam's menus. Holding a controller's Home button works on pads that send it as a Menu key (GuliKit).
 - PC ports (AppImages in ~/Games, like Dusklight) now show up instead of staying on "Starting...", and close cleanly instead of crashing.
 - The Quick Menu no longer freezes or crashes when audio stops answering (after a trip to Desktop Mode).
 - Restart and Power Off warn while an update is still downloading.
 - Wi-Fi shows its real signal on newer kernels.
 - Home (Quick Menu or *hearthctl home*) closes Settings, and *hearthctl status* says when Settings is open.
 - Lighter when idle: a Wii Remote lying still and the screen saver use a fraction of the processor they did.
 - More logs kept (up to 1 GB), so a crash can be looked into afterwards; updates write their progress to their own log.
 - *hearthctl check*, *press* and *screenshot* are more accurate, and *hearthctl windows* shows why an app hasn't reached the screen.
- **Fixes from the second field run** (on this version's first build):
 - Electron apps (VacuumTube, Twitch, Discord) close cleanly from the Quick Menu instead of crashing, and games that take a while to start get a "taking a while to start" note with how to get back home.
 - Settings' sidebar fits every category on the screen; a game picture saved as a palette PNG no longer makes the home screen crash.
 - Game art tries again a few minutes after a boot without network, instead of waiting six hours; Wi-Fi checks no longer stall the screen.
 - *hearthctl screenshot* can't hand back the previous picture, and *hearthctl press* reaches a game on gamescope's second display.
- **People:** add everyone in the household (Settings → People). Hearth then starts on "Who's playing?", with an optional 4-digit PIN each. Each person has their own favorites, Continue row, play times, Steam account and Discord; ROMs, apps and installed games are shared. An admin sets each person's game time, bedtime and locked tiles, and only an admin opens Settings and Desktop Mode. With one person, nothing changes. "Who's playing?" comes back after the PC has slept, and from Quick Menu → System → Switch person; each person can have their own colour scheme.

- **Erasing a drive asks for your password** (Settings → Storage), the one Desktop Mode and *sudo* use. Using a drive as it is doesn't.
- **Game art finds itself**: emulated games without a picture get one from libretro's free thumbnail library, in the background. *hearthctl art* fetches them now; off in Settings → Home screen.
- **GitHub's *gh* command and *tmux* are built in**, so Claude Code on the PC can file field reports, and keeps working if the SSH connection drops (*tmux new -A -s claudes*).

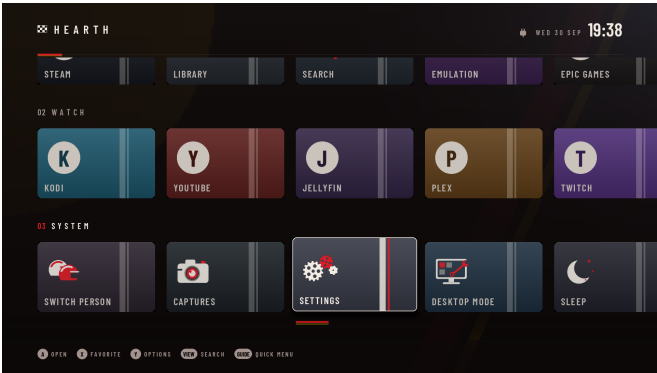
What's working

Area	What it does	Status
Home screen	TV-style rows, liveries, a drawing or icon on every tile, a backdrop from the focused game, controller batteries and network by the clock	Confirmed on the PC
Emulation	ES-DE with a tile per console; emulators tuned for the GPU and TV; game art found automatically	Confirmed on the PC (Dolphin working)
Wii Remote	DolphinBar pointer on the home screen and as a mouse; handed to Dolphin for Wii games	Confirmed on the PC (pointer working)
Quick Resume	Hold Guide to pause a game and pick it up later (up to 3)	Confirmed on the PC
Quick Menu	Guide button panel: audio, per-app volume, Discord, stats, power, updates, screenshots	Confirmed on the PC
Streaming	YouTube (VacuumTube), Twitch (VacuumStream), Kodi, Jellyfin, Plex, Moonlight	Confirmed on the PC
Game stores	Epic Games (Heroic), Battle.net, EA app and Ubisoft Connect (Lutris) tiles; their games in the Library	Shipped
Customising	Favorites with X, move tiles, hide tiles, liveries, sounds	Confirmed on the PC
Search and details	Search every app and game; Y on a game for play time and last played	Confirmed on the PC
Settings	Wi-Fi, static IP and DNS, Bluetooth, audio devices, storage (format and use a new drive)	Confirmed on the PC
Captures	Screenshots from the Quick Menu, a gallery tile, in the screen saver	Confirmed on the PC
Watch next	Shows and films in progress from Jellyfin, Plex and Kodi; resumes and reports back	Shipped
Family	Daily game-time limit, bedtime and locked tiles behind a PIN, per person	Shipped
Privacy	No ads or tracking; a guide to turn off the TV's own tracking, by make	Shipped
TV control (CEC)	TV on and to the right input on wake and on Guide; standby with the PC	Shipped
Performance	Hearth idles under 2% of one core and about 300 MB; hearthctl footprint	Confirmed on the PC
Updates	Automatic, versioned, with releases on GitHub, a what's-new card and rollback	Confirmed on the PC
Phone as a remote	A web page with a d-pad and keyboard, paired with a code on the TV	Not started
People	Who's playing? (at start, after sleep, from the Quick Menu) with optional PINs; own favorites, colour scheme, Continue, Steam and Discord; limits set by an admin	Shipped
Folders	Groups of tiles	Not started (deferred for now)

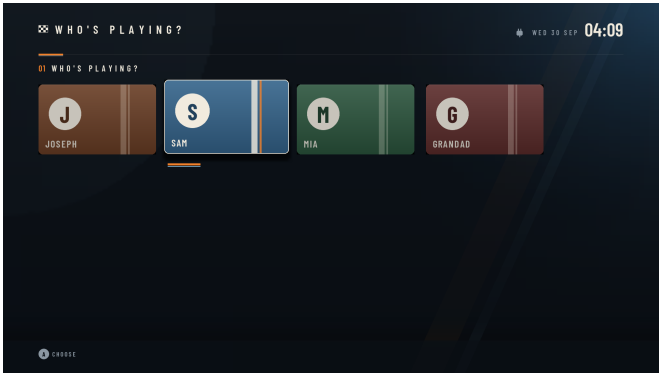
Area	What it does	Status
NVIDIA graphics	A separate image with NVIDIA's driver	Not started (deferred: AMD and Intel first)
Netflix and similar	DRM limits these to low quality on any home-built PC	Not possible

Confirmed means reported working on the real PC; Shipped means built, tested and released, but not yet reported on.

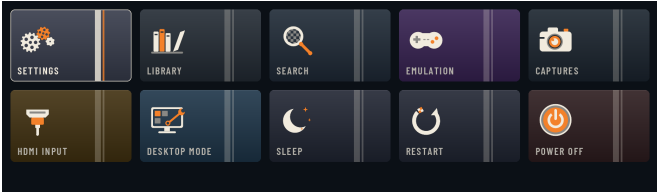
Screenshots



Sam's home screen in the Rosso livery they chose; the household's is Gulf.



Who's playing?: one tile per person, each with an optional PIN.

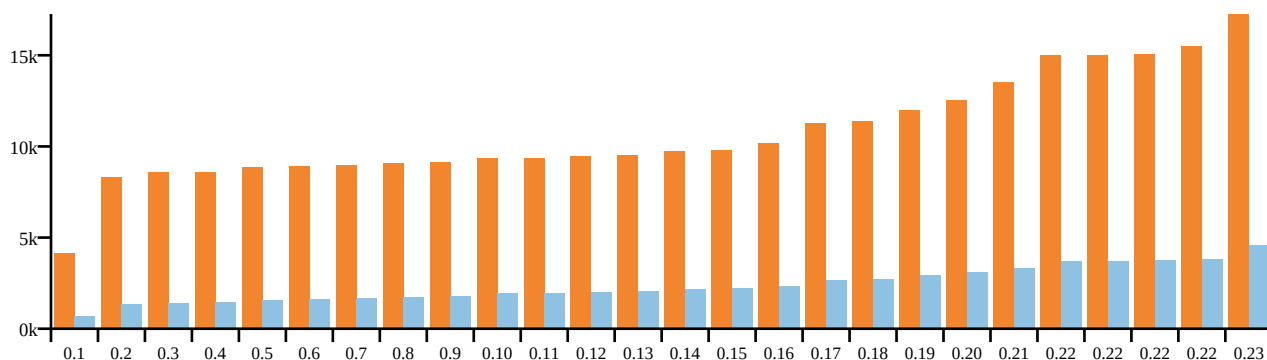


The System tiles, each with its own drawing.

Lines of code by part

Part	Lines
Home screen, Quick Menu and settings (Python)	14,665
System image (scripts, services, config)	1,206
Build and tools	1,390
Tests	6,319

Growth



Lines of code (orange) and automated tests (blue, ×10) at each version.

Open issues

Issue	Where it stands	What's needed
A Steam game started cold from its tile stayed on Steam's spinner (Sekiro)	Seen once in the field tests (#40); the cause isn't known yet	The same game from Steam's own Game Mode, and PROTON_LOG=1
Steam "Switch to Desktop" can still hang	Holding Guide for 4 s always gets you home; the cause isn't known	hearthctl status and hearthctl logs right after it happens
A few things from the field runs still need a look	Two field runs on the PC (0.22.4 and the 0.23.0 build) found 25 problems; all but the Steam cold start above are fixed in 0.23.0	The field tests again on the released 0.23.0 (docs/FIELD_TESTS.md)

Next

- Your phone as a remote, with a keyboard for search and Wi-Fi passwords.
- Fix what turns up on the real PC.
- When wanted: groups of tiles; NVIDIA support.

Every version

Version	Date	Lines of code	Tests	Docs (lines)	Pull requests
0.23.0	2026-09-30	17,261	458	1,990	27
0.22.4	2026-09-30	15,485	381	1,773	27
0.22.2	2026-09-30	15,066	374	1,556	26
0.22.1	2026-09-30	15,008	370	1,548	25
0.22.0	2026-09-29	15,005	369	1,544	24
0.21.0	2026-09-29	13,506	334	1,371	23
0.20.0	2026-09-29	12,533	312	1,260	20
0.19.0	2026-09-29	11,984	291	1,120	19
0.18.0	2026-09-29	11,370	270	1,067	18
0.17.0	2026-09-29	11,288	268	1,034	17
0.16.0	2026-09-29	10,159	233	1,022	16
0.15.0	2026-09-29	9,817	223	971	15
0.14.0	2026-09-29	9,741	218	969	14
0.13.0	2026-09-29	9,507	205	954	13
0.12.0	2026-09-27	9,474	201	954	12
0.11.0	2026-09-27	9,339	195	953	11
0.10.0	2026-09-27	9,326	194	940	10
0.9.0	2026-09-27	9,111	179	906	9
0.8.0	2026-09-27	9,072	173	904	8
0.7.0	2026-09-27	8,970	165	901	7
0.6.0	2026-09-27	8,931	163	901	6
0.5.0	2026-09-27	8,836	157	897	5
0.4.0	2026-09-27	8,598	144	888	4
0.3.0	2026-09-27	8,597	141	886	3
0.2.0	2026-09-27	8,336	132	865	2
0.1.0	2026-09-25	4,155	66	703	1